Sri Lanka Institute of Information Technology



Lab Submission
<Lab sheet 01>

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Discrete Mathematics | IT1160

B.Sc. (Hons) in Information Technology

Part01

01.

In [1]:
```
A = 20
B = 15
C = A+B
print(C)
```

```
35
```

In [81]:
```
A =50
B =12
C =A-B
print(C)
```

```
38
```

In [3]:
```
A =50
B =12
C =A-B
print(C)
```

```
38
```

In [4]:
```
A =8
B =7
C= A*B
print(C)
```

```
56
```

--

In [5]:
```
A =100
B =25
C =A/B
print(C)
```

```
4.0
```

In [6]:
```
A =17
B =5
C =A%B
print(C)
```

```
2
```

02.

In [82]:
```
A = int(input('Enter side01 length: '))
B = int(input('Enter side02 length: '))
C = int(input('Enter side03 length: '))

if A<0 or B<0 or C<0:
    print('Enter a valid input')

if A+B>C and B+C>A and C+A>B:
    print('Valid triangle')

else:
    print('Invalid triangle')
```

```
Enter side01 length: 2
Enter side02 length: 4
Enter side03 length: 5
Valid triangle
```

03.

In [19]:
```
A = int(input('Enter number one: '))
B = int(input('Enter number two: '))
operation = (input('Enter the operation: '))

if operation == '+':
    C=A+B
elif operation == '-':
    C=A-B
elif operation == '*':
    c=A*B
elif operation == '/':
    C=A/B

print('The value is' , C)
```

```
Enter number one: 2
Enter number two: 4
Enter the operation+
The value is {6}
```

04

```
total = 0
for i in range(0,5):
    marks = int(input("Enter marks for the subjects: "))
    total = total+marks

print(total)
```

```
Enter marks for the subjects: 100
Enter marks for the subjects: 100
Enter marks for the subjects: 100
Enter marks for the subjects: 100
Enter marks for the subjects: 100
500
```

05.

```
while True:
    password = input('Enter your password')
    if password == 'python123':
        print('Access granted')
        break;
```

```
Enter your passworddcsdv45
Enter your passwordxvcafac25
Enter your passwordpython123
Access granted
```

Part 02

01.

```
x = int(input('Enter number 01(X): '))
y = int(input('Enter number 02(Y): '))

if(x>y):
    print('X is greater than y')
elif(x!=y):
    print('X is not equal to Y')
elif(x==y):
    print('X is equal to Y')
```

```
Enter number 01(X): 4
Enter number 02(Y): 7
X is not equal to Y
```

02.

```
weight = float(input('Enter the weight(kg): '))
height = float(input('Enter the height(m): '))
BMI = weight/(height*height)

if(BMI<18):
    classification = 'Underweight'
elif(BMI>=18 and BMI<=25):
    classification = 'Normal'
elif(BMI>25):
    classification = 'Overweight'


print(BMI , classification)
```

```
Enter the weight(kg): 54
Enter the height(m): 163
0.0020324438255109336 Underweight
```

03.

```
total = 0
name = input('Enter student\'s name: ')
for i in range(1,4):
    marks = int(input('Enter marks ' + str(i) + ': '))
    total = total+marks

print('Total: ' ,total)
percentage = total/3

if percentage >= 75:
    grade = 'A'
elif percentage >= 60:
    grade = 'B'
elif percentage >= 45:
    grade = 'C'
elif percentage >= 30:
    grade = 'D'
elif percentage < 30:
    grade = 'F'

print('Percentage is: ' ,percentage)
print('Grade' , grade)
```

```
Enter student's name: zc
Enter marks 1: 20
Enter marks 2: 30
Enter marks 3: 10
Total:  60
Percentage is:  20.0
Grade F
```

04.

```
n = int(input('Enter a natural number: '))
sum = 0
for i in range(0,n):

                    sum = sum + number

print('Sum', sum)
```

```
Enter a natural number: 10
Sum 20
```

05.

```
total = 0
while total < 1000:
    price = float(input('Enter the price of items: '))
    total = total+price
print('Checkout complete')
print('Total price: ', total)
```

```
Enter the price of items: 960
Enter the price of items: 120
Checkout complete
Total price:  1080.0
```